CASE STUDY: INTELLIGENT SYSTEMS IN PRODUCTION

REVISED TOPIC

Binary Classification of AOI Alarms into True Defects and False Calls

Second check of AOI results with supervised machine learning (Siemens production data)

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Team 1 · Guided by Dr. Hamidreza Heidari · 23 October 2026 · github.com/bodhi584/aoi-decision-gate

INTRODUCTION

Too many false alarms after automated inspection

98 %

of the boards flagged by AOI are false calls

Each one is checked again by hand at the manual inspection station.

Source: Siemens SMT line, Bavaria · 39,742 flagged boards, 132 days · Pfab et al. (2024), Data in Brief 53, 110110

Based on the lecture example (p. 18)

“Siemens – Quality Control and Defect Detection”. Key motivation: fewer manual re-inspections of false calls, while true defects are still caught.

A current problem: Siemens and NVIDIA, Erlangen, 21 Oct 2026

“AI second check of AOI results”: real defect or false call? Target: about 40 % fewer false calls.

Our goal

Classify each AOI alarm as true defect or false call, so that fewer false calls go to manual inspection while at most 1 % of true defects are missed.

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ONE GOAL: FEWER FALSE CALLS TO MANUAL INSPECTION, AT MOST 1 % OF TRUE DEFECTS MISSED

Input → System → Output

INPUT

AOI alarm

one flagged board: about 70 AOI measurements per flagged solder joint

component type, AOI error code

SYSTEM (OURS)

Binary classification

supervised machine learning, trained on past manual-inspection results

classification threshold: recall ≥ 99 %

OUTPUT: IS THE AOI ALARM REAL?

False call

→ skip manual inspection

True defect

→ manual inspection, repair

Key factors

False calls

each one costs operator time at the manual station

Missed defects

a missed defective board reaches the customer

Changes over time

the line changes; a model must keep up

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PROJECT SCOPE

What we will and will not do

Aim

This project aims to reduce false calls after automated optical inspection: a binary classifier decides for each AOI alarm whether it is a true defect or a false call, while at most 1 % of true defects are missed.

Includes

- Data preprocessing: Siemens AOI data
- Training and comparing three classification models
- Day-by-day simulation on recorded data
- Evaluation: recall, specificity, confusion matrix

Out of scope (current phase)

- Real hardware deployment
- Integration with physical AOI machines
- Changing the AOI inspection itself (camera, limits)

Conditions and limitations

- One Siemens line, 132 days
- Measurements anonymised (no physical names)
- Labels from operators may contain errors
- Licence: non-commercial (course use)

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What the project will achieve, and how we measure it

1. Reduce false calls: train three classification models (logistic regression, random forest, neural network) for the same binary task.
2. Keep quality: set the classification threshold so that recall of true defects is at least 99 %.
3. Test as in operation: day-by-day simulation on the recorded Siemens data, models trained on earlier days only.
4. Compare the three models and identify the most suitable one.

Requirements (positive class = true defect)

≥ 99 % recall: true defects classified as true defect (≤ 1 % missed)

≥ 20 % specificity: false calls classified as false call = manual inspections saved (baseline 6.4 %)

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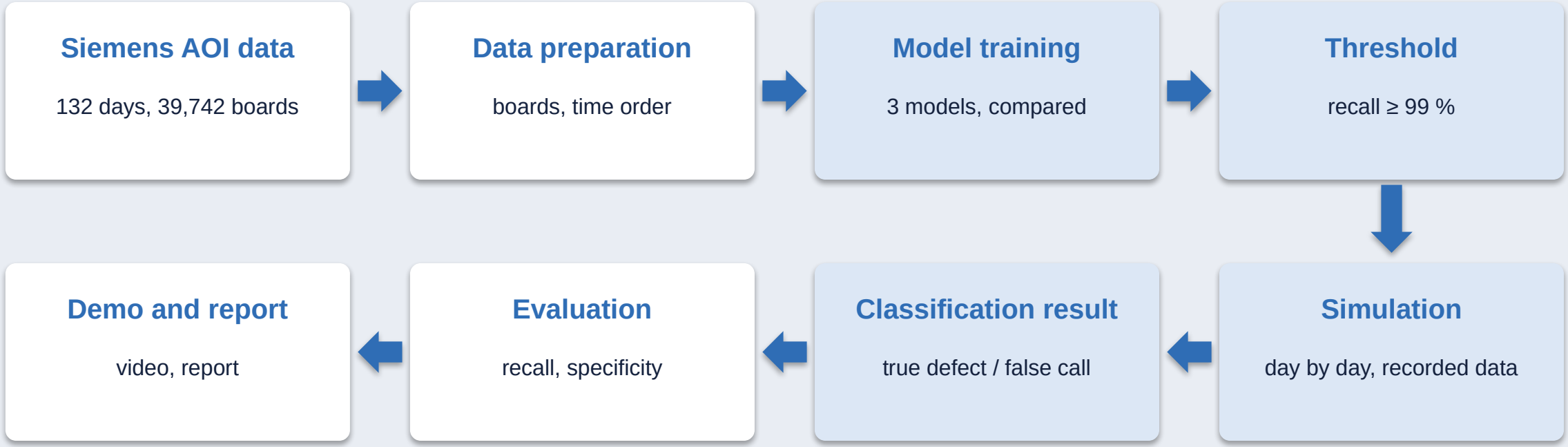
TECHNOLOGY STACK

Data science tools from the lecture (p. 22)

Process	Tool / Framework	Purpose
Data preparation	Python, pandas	Load the Siemens AOI data, group joints into boards, keep time order
Model training	scikit-learn	Logistic regression, random forest, neural network (one hidden layer)
Simulation	Python	Day-by-day simulation on recorded production days (walk-forward)
Evaluation	scikit-learn, pandas	Recall, specificity, confusion matrix
Visualization	matplotlib, Jupyter	Graphical result representation, step-by-step notebook
Code sharing and tests	GitHub, pytest	Repository, tests on every change

PROJECT FLOW CHART

From the Siemens data to the KPIs



Blue boxes: the system we build. White boxes: data and project steps.

WORKING METHODOLOGY

CRISP-DM in four steps

Step 1 · Data preparation

Collected Siemens AOI data: flagged boards, defective & false call

Grouped solder joints into boards; encoded categories

Split data in time order into train, validate, test sets

Step 2 · Model development

Trained logistic regression as baseline

Trained random forest and a neural network (one hidden layer)

Set the classification threshold: recall ≥ 99 %

Step 3 · Simulation integration

Simulated operation day by day on the recorded days

Each day classified by a model trained on earlier days only

Boards classified as false call virtually skip manual inspection

Step 4 · Evaluation and comparison

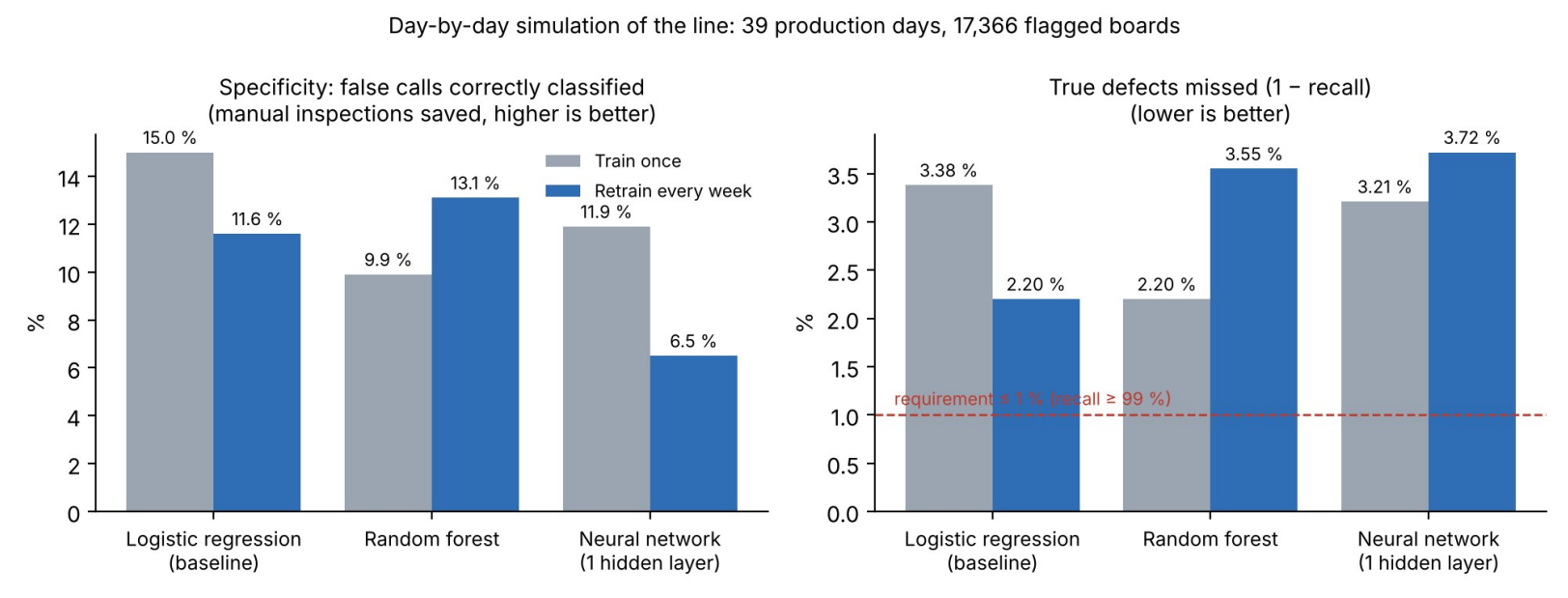
Analyzed recall, specificity and the confusion matrix

Compared the three models to identify the best-performing one

Error analysis; demo video and report

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Day-by-day simulation on recorded data: three models compared



What it shows

Specificity today: 6.5–15.0 % (false calls correctly classified).

Recall today: 96.3–97.8 % (2.2–3.7 % of true defects missed): below the 99 % requirement.

Next: recall ≥ 99 % with specificity ≥ 20 %.

BILL OF MATERIALS (BOM)

What the project needs

Component	Specification	Quantity	Cost
Data set	Siemens AOI data, Mendeley (CC BY-NC 3.0), 333 MB	1	Free
Computer	Normal laptop or Google Colab (no graphics card needed)	4	Free
Software	Python and open-source libraries	–	Free
Code hosting	GitHub	1 repository	Free

Total cost: 0 €

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EXPECTED OUTCOMES

What we hand in

- ✓ Successfully trained binary classifier for AOI alarms (Python software prototype)
- ✓ Day-by-day simulation on the recorded data, with recall and specificity
- ✓ Comparative performance metrics for the three classification models
- ✓ Identification of the most suitable model, with limits and error analysis
- ✓ Demo video, run instructions, report, public GitHub repository

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SAFETY & ETHICAL CHECKLIST

Keeping it safe and realistic

- ✓ Simulation-only environment; no physical machinery risk
- ✓ Dataset is anonymized by Siemens (no personal data); used under its licence
- ✓ Ethical principles followed: transparency, fairness, accountability
- ✓ Avoid dataset bias (time-ordered split) and ensure reproducibility of results
- ✓ Safety first in the design: recall of true defects ≥ 99 %

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FUTURE WORK

From the public data set to a real line

1 · Industry partner

An electronics plant shares AOI and manual-inspection records.

2 · AOI interface

Read each board's AOI result from the machine; send the classification to the manual inspection station.

3 · Shadow mode

The classifier runs next to the operators and only logs its results.

4 · Approval

Quality engineers compare the logs before any board skips manual inspection.

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Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830.

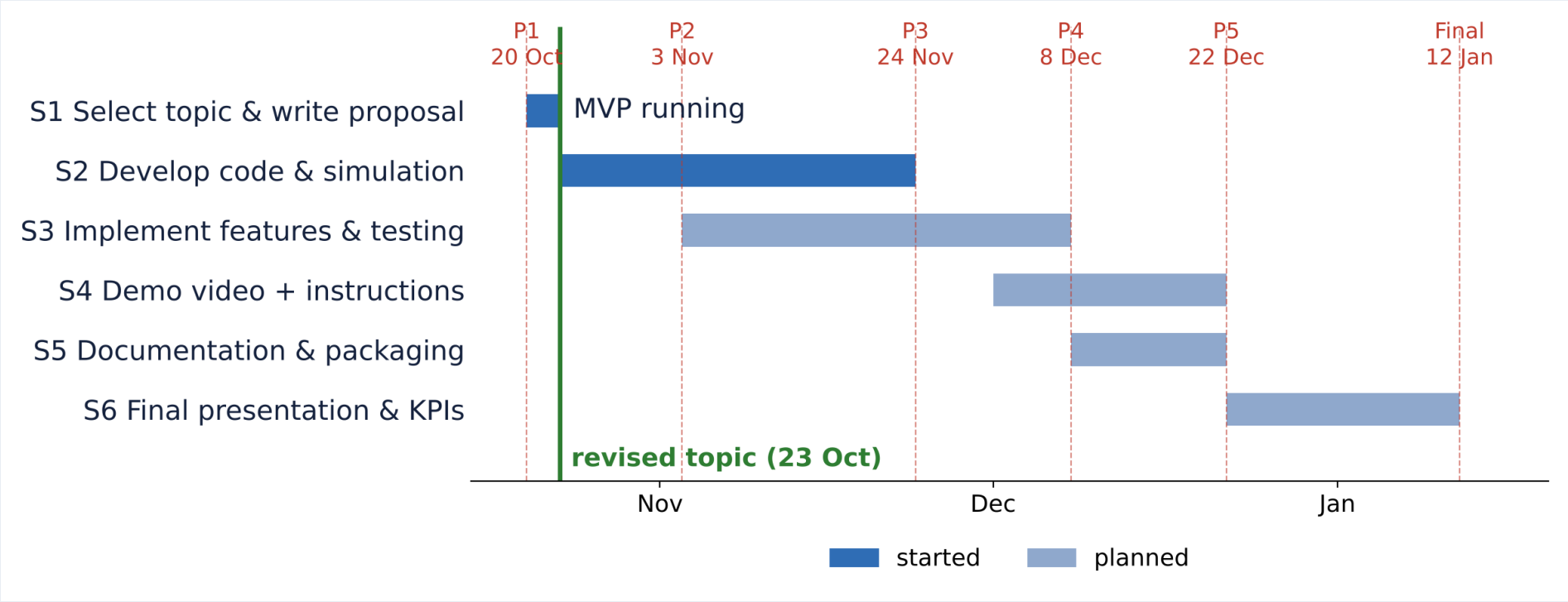
Chapman, P. et al. (2000). CRISP-DM 1.0: Step-by-step data mining guide. SPSS.

Lecture slides, Case Study Intelligent Systems in Production, p. 8, 12, 18, 22, 33.

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MILESTONES

Software & Simulation project, S1–S6 (lecture p. 8, 33)



Repository link

github.com/bodhi584/aoi-decision-gate

Thank you

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